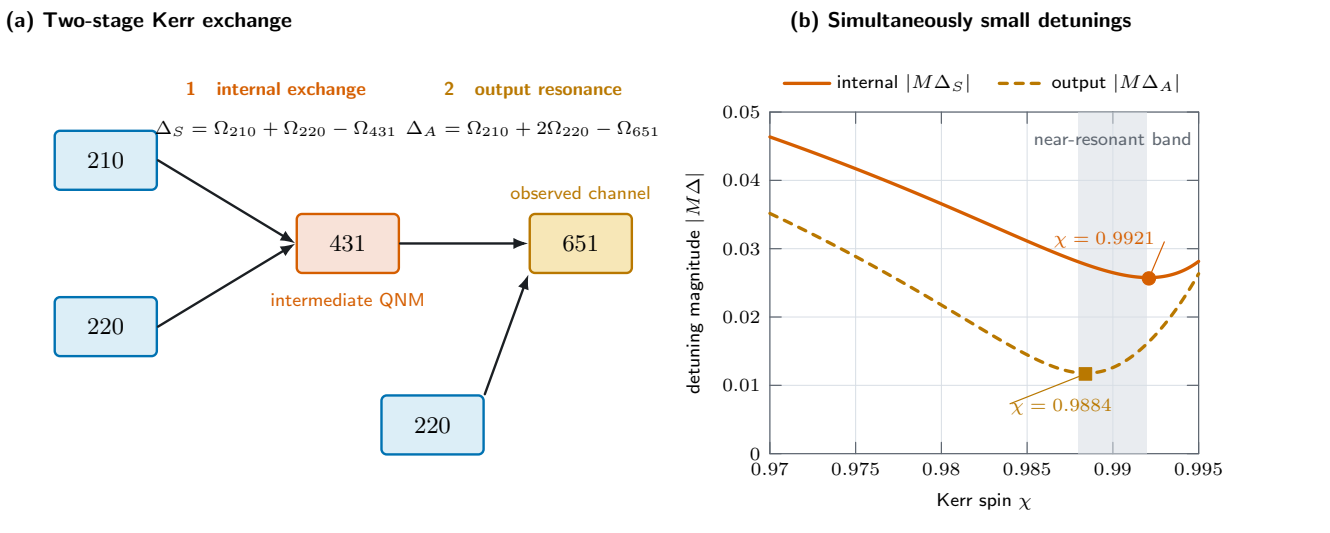

(a) Two-stage Kerr exchange
1 internal exchange
2 output resonance
$\Delta_S = \Omega_{210} + \Omega_{220} - \Omega_{431}$ $\Delta_A = \Omega_{210} + 2\Omega_{220} - \Omega_{651}$
210
220
431
intermediate QNM
observed channel
651
220
(b) Simultaneously small detunings
internal $|M\Delta_S|$
output $|M\Delta_A|$
near-resonant band
$\chi = 0.9921$
$\chi = 0.9884$
detuning magnitude $|M\Delta|$
Kerr spin χ
0.05
0.04
0.03
0.02
0.01
0
0.97
0.975
0.98
0.985
0.99
0.995